MILITARY SPECIFICATION

DISTRIBUTOR, BITUMINOUS MATERIAL, TRUCK MOUNTED, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the U.S. Army Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a commercial truck-mounted, bituminous distributor.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-PD-9156	- Truck and Truck Tractor Commercial, Diesel Engine Driven 55000 to 75000 GVWR.

FSC 3895

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

(Copies of specifications and standard required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

DEPARTMENT OF LABOR

Occupational Safety Hazard Act.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to USA Mobility Equipment Research and Development Command, ATTN: DRXFB-H, Fort Belvoir, VA 22060.)

3. REQUIREMENTS

3.1 Description. This specification covers a truck-mounted, bituminous distributor with a hydrostatically driven bituminous pump. The distributor shall be the current commercial production model of the standard product of the manufacturer and, except for product improvements

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which have been incorporated in the commercial production units, shall have been marketed for at least 1 year prior to the opening of the technical proposal or, in the case of a recently introduced item, the manufacturer shall submit sufficient valid industrial operational/test data to demonstrate the acceptability of the item. In addition to the above the distributor shall be complete with all components furnished as standard equipment that are customarily offered with a distributor of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the distributor to function reliably and efficiently in a sustained operation.

3.2 Function. The distributor shall be capable of the following functions:

(a) Pump bitumen from an external source into the bitumen tank without bitumen entering the spraybar.
(b) Pump bitumen from an external source to another external source without bitumen entering the tank or the spraybar.
(c) Pump bitumen from the bitumen tank back to the bitumen tank without the bitumen entering the spraybar.
(d) Pump bitumen from the bitumen tank through the spraybar back to the bitumen tank.
(e) Pump bitumen out of the bitumen tank to an external source without the bitumen entering the spraybar.
(f) Uniformaly apply bitumen at rates of from 0.1 to not less than 1 gallon per square yard and at any rate in between without streaking or puddling.
(g) Pump bitumen out of the bitumen tank to the hand-spray attachment.
(h) Withdraw bitumen from the spraybar and all piping and return to the bitumen tank.
(i) Pump flushing oil through all piping and the spraybar without the flushing oil entering the bitumen tank.
(j) Withdraw bitumen from the hand-spray attachment or blow-out bitumen from hand-spray attachment.

3.3 Bituminous tank. The tank shall be insulated and, unless otherwise specified, shall have a capacity of not less than 1500 gallons plus not less than 4 percent for contents expansion. When specified (see 6.2), the tank shall have a capacity of not less than 2000 gallons plus not less than 4 percent for contents expansion. An overflow pipe shall be provided to direct any bitumen overflow to the ground. The tank shall be provided with baffle plates adequate to withstand surge pressures normally encountered during operation. A pencil-type thermometer shall be provided for the tank.

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3.4 Heating system. The heating system shall consist of two U-tube type heating flues and two low pressure air atomizing burners, fuel pump, fuel tank with a capacity of not less than 20 gallons and blower. Each burner shall have a rating of not less than 500,000 Btu/hr and shall be capable of operating on standard Army diesel fuel. When a 2000 gallon tank is specified (see 6.2), each burner shall have a rating of not less than 900,000 Btu/hr. The burner system fuel pump shall be connected to the power source so that it will operate only when the blower is operating. The blower operation shall be independent of the bituminous pump.

3.5 Spraying system. The spraying system shall include a positive-displacement rotary type bituminous pump having a capacity of not less than 375 gallons per minute. Provisions shall be made to permit heating of the pump with the portable burner. The bituminous pump in conjunction with the power take off and hydrostatic transmission (see 3.6) shall provide bitumen to the spray bar that will produce application rates of from 0.1 gal/sq yd to not less than 1.0 gal/sq yd through the 24-foot spraybar. The actual rate of application shall be within 10 percent of the intended rate. The spraybar shall be of the full circulating type and shall consist of a center section approximately 8 feet in length plus turn-up type extension ends. Each spray nozzle shall be capable of independent operation or simultaneous operation with all nozzles. Remote type controls shall be provided in the truck cab and at the rear operator's platform to (a) turn bitumen flow from the spraybar on or off, (b) shift spraybar laterally (c) vertically raise or turn up the spraybar as applicable (distributors having the turn-up feature shall be provided with mechanical means to vertically adjust spray height when desired). Spraybar extensions to allow spraying to 24 feet in width shall be provided. Each extension section shall be not greater than 2 feet in length. Lateral movement shall be not less than 8 inches each way from center. The spraybar shall be provided with a break-away feature to protect it from damage in the event an obstruction is hit by the spraybar.

3.6 Hydrostatic transmission. The bituminous pump shall be driven by a hydrostatic transmission. The PTO control shall be located so as to be engaged/disengaged by the operator in the truck cab. The control that varies the speed of the hydrostatic transmission shall be located in the truck cab with override controls provided at the rear platform.

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3.7 Attachments. The following attachments shall be provided and shall be the latest model of the standard product of the offeror:

(a) Portable burner for heating bitumen pump and piping.
(b) Two 15-foot sections of filling hose equipped with quick-disconnect fittings.
(c) Bituminous tank volume gage.
(d) Bitumeter wheel.
(e) Control in the cab to raise and lower the bitumeter wheel.
(f) Box for storage of spraybar extension sections.
(g) A hand-spray attachment with not less than 25 feet of hose.
(h) Signal gong for rear operator-driver communication.
(i) Distribution charts, encased in a transparent protective material.
(j) Strainers (manhole, asphalt fill line, asphalt pump).
(k) Pump tachometer, calibrated in gallons per minute, in truck cab.
(l) Speed and distance indicator reading in feet per minute, located in truck cab.
(m) Toolbox with hasp.

3.8 Truck chassis. The distributor shall be mounted either on a Government-furnished M809 5-ton truck chassis or a Government-furnished commercial truck chassis in accordance with MIL-T-PD-9156 (see 6.2). The alinement between truck chassis and distributor shall be maintained, and the mounting devices shall not be permanently deformed or broken when the fully loaded, truck-mounted distributor is subjected to travel over paved and gravel roads and subjected to sudden stops when traveling at forward speeds up to 10 mph or at maximum, high-range reverse speeds.

3.9 Lubrication. The truck mounted distributor shall be capable of operating using standard military lubricants without deleterious effects due to lubrication. All components requiring lubrication shall be provided with means of lubrication. Initial servicing of each truck mounted distributor shall be with military lubricants.

3.10 Maintenance characteristics. The truck mounted distributor design, together with the accessories, shall permit ready access to all items requiring periodic maintenance service in the field. Periodic maintenance will be accomplished with the use of conventional general purpose tools normally associated with equipment of this type. The adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the distributor.

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3.11 Lifting and tiedown. Procedures for lifting and tiedown of the complete truck mounted distributor shall be provided. These procedures shall be included in the appropriate manuals for the truck mounted distributor. Instruction plates complete with diagrams and instructions for use of the lifting and tiedown procedures shall be permanently affixed to the distributor. Each lifting or tiedown point shall be identified by a stencil marking.

3.12 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded, or insulated to the extent necessary to eliminate the hazard. External equipment including catwalks, handrails, fenders and bumpers will be designed for durability and operator safety. All catwalks shall be provided with anti-skid surfacing. All platform steps, and handholds shall be of such design to permit unhampered and nonhazardous access to and from the distributor. If the noise level in the area occupied by the operator exceeds 85 dbA, as measured in accordance with SAE J919, a warning plate containing the following shall be permanently affixed to the distributor in a conspicuous protected location:

"Warning, Ear Protection Required When Distributor Is Operating."

The plate shall have a yellow background with black lettering and shall be of corrosion-resistant material.

3.13 Treatment and painting. The parts of the truck mounted distributor normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. The final color shall be forest green in accordance with MIL-T-704.

3.14 Marking. The US Army marking and registration numbers shall be as specified in the solicitation (see 6.2). Marking and registration shall be in accordance with MIL-STD-1223, color black.

3.15 Government-furnished equipment (GFE). The Government shall furnish:

(a) Logistics Support Plan.
(b) Initial Prescribed Load List and Initial Authorized Stock List.
(c) Truck Chassis.
(d) Vehicle Classification Date Sheet.

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3.16 Basic issue items and commercial manuals. Each truck mounted distributor shall be overpacked with:

(a) The items specified in the Basic Issue Items List.
(b) Two copies of the Government-approved commercial Operator, Parts, Maintenance and Overhaul Manuals and Lubrication Order.

3.17 Identification and instruction plates. The manufacturer shall provide all identification and instruction plates normally furnished. A data plate shall be provided stating the manufacturer, model, National Stock Number, and date of acceptance by the Government. The plates shall be permanently affixed to the truck mounted distributor in a conspicuous location.

3.18 Technical Information Package (TIP). The information in the TIP shall be completed depicting the complete truck mounted distributor and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used. The TIP shall be submitted with all bids and proposals.

3.19 Truck vehicle classification. The manufacturer shall complete and submit the Government-furnished "Vehicle Classification Data Sheet" at the First Article Evaluation (see 4.2).

3.20 Technical publications. Such technical publications are as specified shall be furnished (see 6.2).

3.21 Special tools. Any special tools required for operation and operator maintenance of the distributor, except chassis, shall be provided with each truck mounted distributor.

3.22 Certification. Prior to the First Article Evaluation (see 4.2), the manufacturer shall provide certification that the truck mounted distributor conforms to the following:

(a) Applicable Department of Labor (OSHA) regulations.
(b) That the burners have the capacity specified in 3.4.
(c) Department of Transportation, Federal Motor Vehicle Safety Standards and Federal Motor Carrier Safety Regulations (except for FMVSS 121).
(d) That the distributor will satisfactorily operate on military lubricants as specified in 3.9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article evaluation. Within the time frame specified (see 6.2), the manufacturer shall make a truck-mounted distributor available for first article evaluation and demonstration. This distributor and its components shall be compared with this document, and the manufacturer's completed TIP to verify conformance to this specification. In addition, the truck-mounted distributor shall be subjected to a demonstration as specified herein. Nonconformance to any requirement contained herein shall be cause for rejection of the truck mounted distributor by the Government. The manufacturer shall submit a report within 30 days after completion of the evaluation. The report shall include a detailed description of the conformance examination and demonstration performed, deficiencies encountered, and actions taken to correct the deficiencies.

4.2.1 Demonstration. A demonstration of the truck mounted distributor shall be performed as specified herein. Failure of any demonstration shall be cause for refusal to continue acceptance of the distributor by the Government until evidence has been provided that corrective action by the manufacturer has been taken to eliminate the deficiencies.

4.2.1.1 Functional. Demonstrate the distributor's ability to perform the functions listed in 3.2 (except Item (f)). In addition, adjust the spraybar laterally and vertically (or turn up) using the controls provided. Failure to perform any function or malfunction of any control shall be cause for rejection.

4.2.1.2 Distribution. Material used for this demonstration shall be any bituminous material normally used in distribution operations heated to recommended spraying temperature. Operate the distributor to apply bitumen through the 12-foot spraybar at rates of 0.1, 0.4, and 0.8 gallons per square yard and through the 24-foot spraybar at a rate of not less than 1.0 gallon per square yard. The above application rates shall be applied over a predetermined distance of not less than 100 feet for each rate. The following sequence shall be followed for each rate:

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(1) Weigh the loaded distributor.
(2) Apply the bitumen over the predetermined distance at the required rate of application. (Travel speed, pump setting shall be obtained from the application chart).
(3) Reweigh the distributor.
(4) Determine the amount of bitumen applied by difference in weight before and after test and convert to gallons. (Use applicable certified specific gravity vs temperature curve.)
(5) Compute rate of application in gallons per square yard by utilizing the amount of bitumen applied, travel distance and spraybar width.

Any rate not within 10 percent of the intended rate shall be cause for rejection.

4.2.1.3 Axle loads. Determine, by weighing, the loads on the front axle and rear tandem with the distributor and all tanks and reservoirs fully loaded. Record the results.

4.2.1.4 Turning capability. Demonstrate vehicle clearance axle in accordance with SAE J695. Field Test Procedures.

4.2.1.5 Braking capability. Demonstrate the brake capability of the vehicle to determine conformance to Department of Transportation requirements.

4.2.1.6 Mounting.

(a) Accelerate the fully loaded distributor forward from rest to 10 mph as quickly as possible, then apply brakes to stop suddenly. When forward motion has stopped, immediately accelerate to 10 mph. Repeat the acceleration-forward and sudden-stop procedure 10 consecutive times.
(b) Accelerate backward from rest to maximum reverse high range speed as quickly as possible, then apply the brakes to stop suddenly. When reverse motion has stopped, immediately accelerate to maximum high range reverse speed. Repeat the reverse-acceleration and sudden-stop procedure 10 consecutive times.
(c) Operate the distributor fully loaded with bitumen on primary and secondary highways for 250 miles each respectively.

Any breakage or permanent deformation of the mounting devices or any misalinement between distributor and chassis shall be cause for rejectio

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4.3 Run-in. Each distributor shall be run-in in accordance with the manufacturer's standard practice.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete distributor and all items issued with the distributor shall be suitably preserved and packed for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the distributor in operation. Marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract or purchase order number, and appropriate address markings on each unpacked item or shipping container. The marking shall be applied with a waterproof material, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The truck mounted distributor is intended for distributing liquid bitumen for road and airfield construction.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Items listed in the Basic Issue Item List (see 3.16).
(c) Time frame for issuing and submitting the TIP (see 3.18).
(d) US Army marking and registration numbers and location (see 3.14).
(e) Time frame for submitting the first article for demonstration (see 4.2).
(f) The truck chassis to be used (see 3.8).
(g) When a 2000 gallon tank is required (see 3.3).
(h) Type and number of Technical Publications to be furnished (see 3.17).
(i) Time frame for completing and submitting the technical Information Package (see 3.18).

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6.3 Government-furnished property. The contracting officer shall arrange to furnish the property listed in 3.15.

6.4 Publications. The contracting officer will determine that the manufacturer's operating, maintenance and parts manuals and lubrication instructions are approved by the Government prior to accepting the first article.

Custodian:
Army - ME

Review activity:
Army - AT

Preparing activity:
Army - ME

Project 3895-A161

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-D-52916(ME) Distributor, Bituminous Material, Truck Mounted, Commercial Construction Equipment (CCE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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